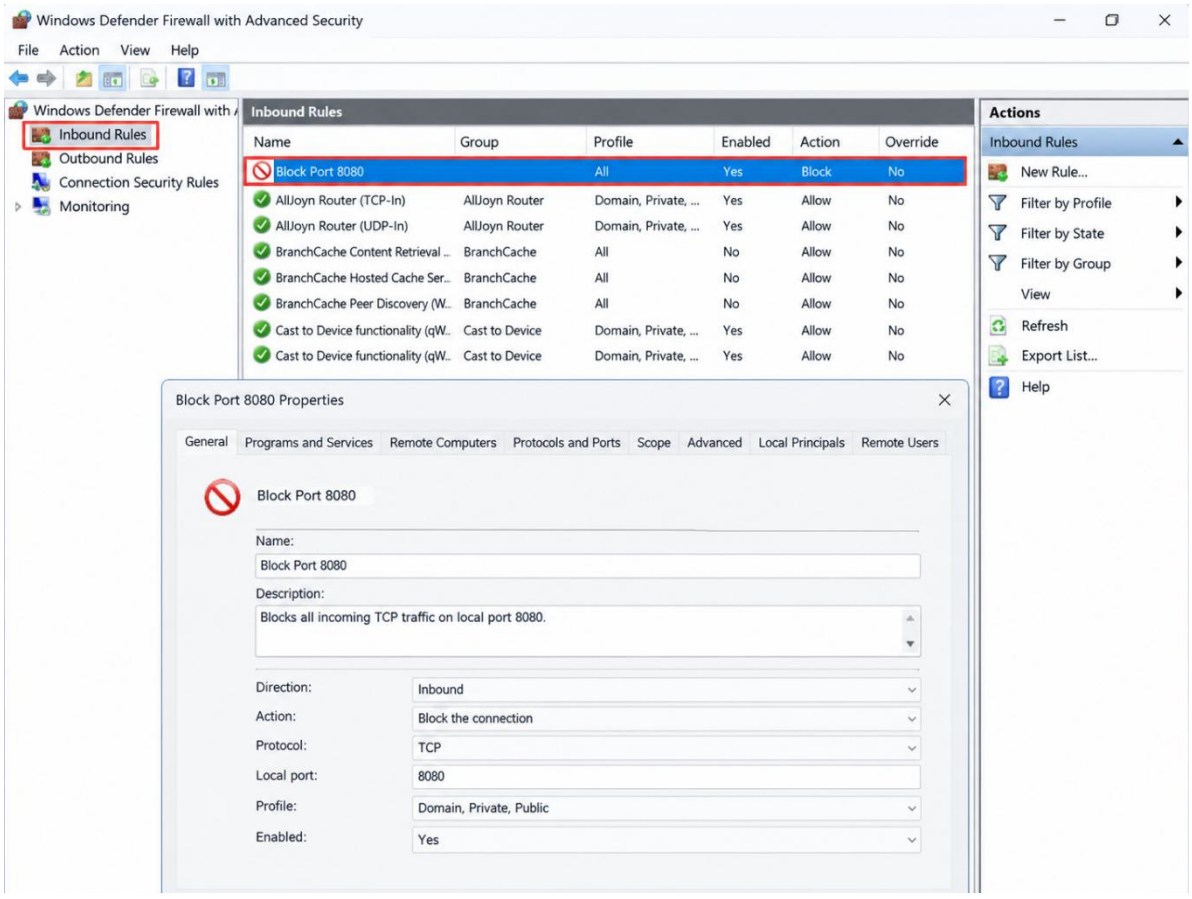
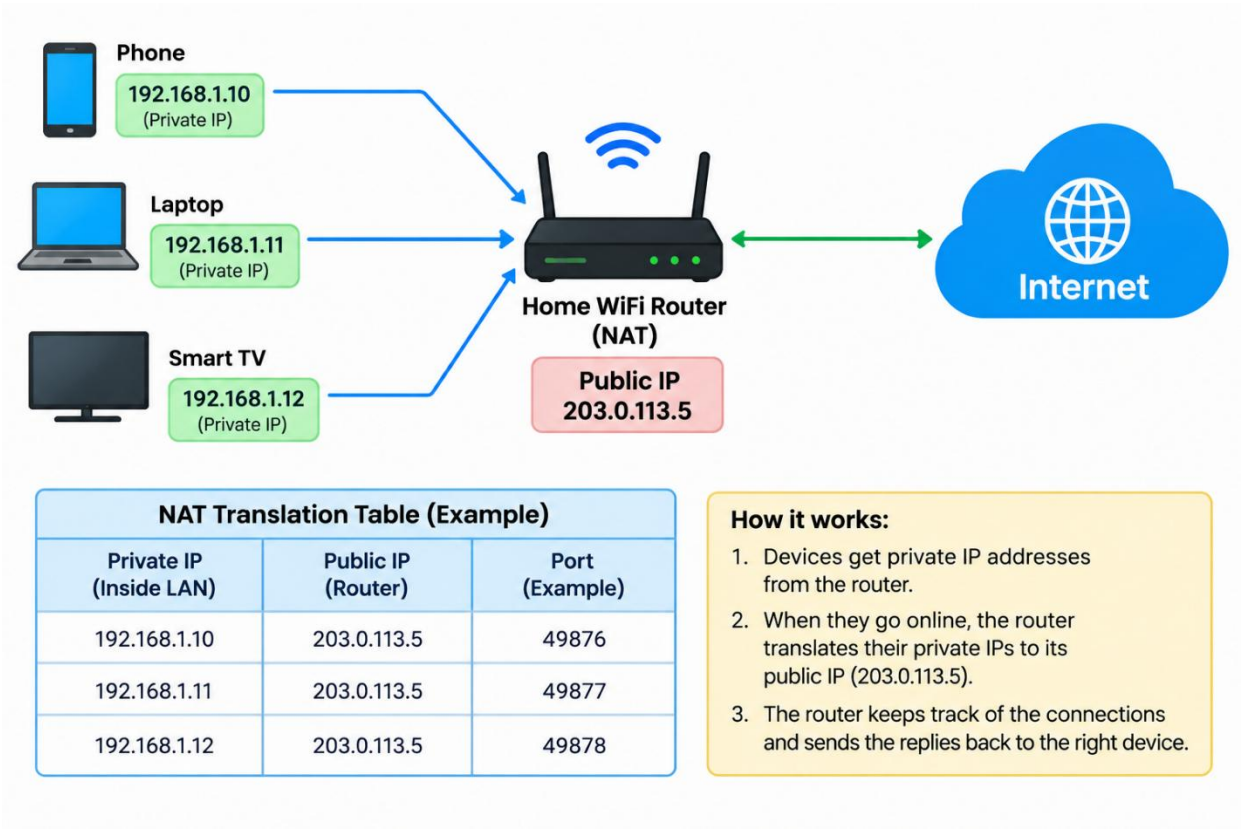
Assignment – 2.2

Secure networks with firewalls NAT, port filtering, packet filtering, and other methods

Q - 1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.



Q - 2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access. Hint: Use any drawing tool or even hand-draw and click a photo.

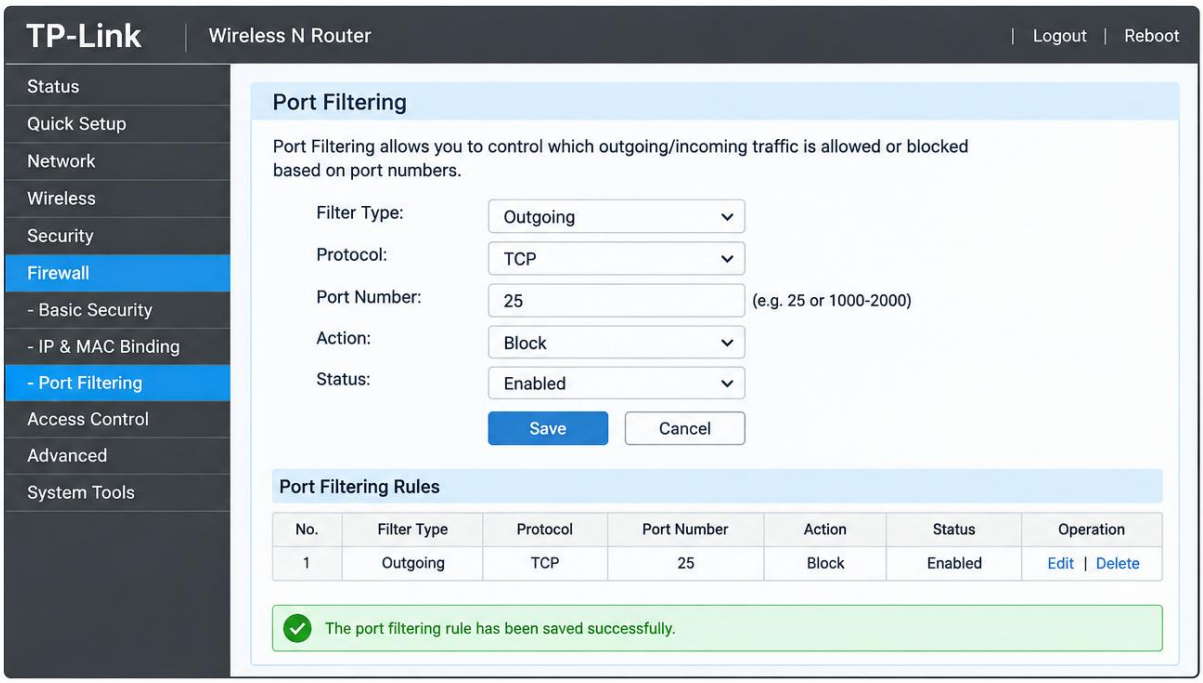


NAT Translation Table (Example)		
Private IP (Inside LAN)	Public IP (Router)	Port (Example)
192.168.1.10	203.0.113.5	49876
192.168.1.11	203.0.113.5	49877
192.168.1.12	203.0.113.5	49878

How it works:

1. Devices get private IP addresses from the router.
2. When they go online, the router translates their private IPs to its public IP (203.0.113.5).
3. The router keeps track of the connections and sends the replies back to the right device.

Q - 3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.



➢ **Explanation:**

Blocking outgoing traffic on port 25 (SMTP) prevents devices on the network from sending emails directly to external mail servers. This helps prevent spam emails from being sent by malware or compromised devices on your network.

Q - 4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53. Hint: Focus on source/destination IPs and ports in the rule set.

➢ Packet Filtering Rules – Allowed or Blocked:

The packets can be checked by comparing their source IP, destination IP, and destination port with the firewall rules provided by the trainer.

No.	Packet	Protocol	Destination Port	Result
1	192.168.1.5 → 172.217.166.78	HTTP	80	Allowed*
2	10.0.0.10 → 34.201.23.1	SSH	22	Blocked*
3	192.168.1.7 → 8.8.8.8	DNS	53	Allowed*

➢ Explanation:

1. **HTTP (Port 80):** Allowed if the firewall permits HTTP traffic.
2. **SSH (Port 22):** Blocked if the firewall denies SSH traffic.
3. **DNS (Port 53):** Allowed if the firewall permits DNS queries.

Q - 5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

- **Three Ways Instagram or Zomato Use Firewalls and Packet Filtering:**

  1. **Blocking unauthorized access:** Firewalls block unknown users from accessing Instagram or Zomato servers.

  2. **Filtering ports:** Packet filtering allows safe traffic, such as HTTPS (port 443), and blocks unwanted traffic.

  3. **Preventing attacks:** Firewalls help block suspicious traffic and protect servers from hackers and network attacks.